

IB MATH AA SL

Lesson 8

The Binomial Theorem

Number & Algebra · 60-minute lesson

Syllabus SL 1.9 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. Expand $(a + b)^2$
2. Work out $4!$ (factorial)
3. Pascal's triangle: write row 4 (start 1, 4, ...)

Answers (click to reveal)

1. $a^2 + 2ab + b^2$
2. 24
3. 1, 4, 6, 4, 1

What we're learning today

LEARNING OBJECTIVES

- 1 Use**
Pascal's triangle / binomial coefficients.
- 2 Expand**
 $(a + b)^n$ for positive integer n .
- 3 Find**
a specific term in an expansion.
- 4 Apply**
the nCr formula for the coefficients.

SUCCESS CRITERIA — I CAN...

- ✓
write binomial coefficients using nCr
- ✓
expand a bracket with the binomial theorem
- ✓
pick out a particular term without full expansion
- ✓
evaluate the coefficient of a given power

The binomial theorem

THE FORMULA

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

nCr

= $n! / (r!(n-r)!)$, on your GDC too

Powers

a's power falls, b's power rises; they add to n

COEFFICIENTS

Row n of Pascal's triangle gives the nCr values.

(n=4)

1, 4, 6, 4, 1

GDC

use nCr for large n.

Full expansion

QUESTION

Expand $(1 + 2x)^4$.

SOLUTION

Coefficients (row 4): 1, 4, 6, 4, 1

Apply with $a = 1$, $b = 2x$:

$$1 + 4(2x) + 6(2x)^2 + 4(2x)^3 + (2x)^4$$

Simplify each term:

$$= 1 + 8x + 24x^2 + 32x^3 + 16x^4$$

Your turn — expand

ANSWER IN YOUR BOOK

1. Expand $(1 + x)^3$
2. Expand $(2 + x)^3$
3. Write the coefficients for $n = 5$

Answers (click to reveal)

1. $1+3x+3x^2+x^3$
2. $8+12x+6x^2+x^3$
3. 1,5,10,10,5,1

Finding one term

THE GENERAL TERM

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

Choose r so the power of x is the one you want.

No need to expand the whole bracket.

STRATEGY

1.
write the general term
2.
set the power of x equal to the target
3.
solve for r , then evaluate

A specific term

QUESTION

Find the coefficient of x^2 in the expansion of $(1 + 2x)^4$.

SOLUTION

General term: $C(4,r) \cdot 1^{4-r} \cdot (2x)^r$

For x^2 we need $r = 2$:

$$\binom{4}{2} (2x)^2 = 6 \times 4x^2$$

So the coefficient is:

$$= 24$$

Exam-style question

Consider the expansion of $(3 + x)^5$.

- (a) Write down the number of terms. [1]
- (b) Find the term in x^2 . [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) 6 terms (A1)
- (b) $C(5,2) \cdot 3^3 \cdot x^2 = 10 \cdot 27 \cdot x^2$ (M1)(A1) = $270x^2$ (A1)

Common mistakes



Index on the bracket

$(2x)^2 = 4x^2$, not $2x^2$ — square everything in the bracket.



nCr order

$C(n,r)$ uses r for the SECOND term's power.



Counting terms

$(a+b)^n$ has $n+1$ terms, not n .



Forgetting coefficients

Don't drop the Pascal/ nCr numbers.

Mixed practice

QUESTIONS

1. Expand $(1 + x)^4$
2. Find the coefficient of x^3 in $(1 + x)^5$
3. Expand $(2 - x)^3$
4. Find the term in x^2 of $(1 + 3x)^4$

Answers (click to reveal)

1. $1+4x+6x^2+4x^3+x^4$
2. $C(5,3)=10$
3. $8-12x+6x^2-x^3$
4. $C(4,2) \cdot (3x)^2 = 6 \cdot 9x^2 = 54x^2$

Plenary & exit ticket

KEY FORMULAS

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

EXIT TICKET — before you leave

1. Coefficients for $n = 3$?
2. How many terms in $(a+b)^6$?
3. Coefficient of x^2 in $(1+x)^4$?

Answers (click to reveal)

1. 1, 3, 3, 1
2. 7
3. 6

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

The binomial theorem.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 9

Proof & deductive reasoning — writing a clear mathematical argument.

$2n+1$ is odd